



Bootling linux on ESP32s3

Sources:

- <https://github.com/jcmvbkbc/xtensa-dynconfig/tree/original> [<https://github.com/jcmvbkbc/xtensa-dynconfig/tree/original>]
- <https://github.com/jcmvbkbc/config-esp32s3> [<https://github.com/jcmvbkbc/config-esp32s3>]
- <https://github.com/jcmvbkbc/esp-idf/tree/linux-5.0.1> [<https://github.com/jcmvbkbc/esp-idf/tree/linux-5.0.1>]
- <https://github.com/jcmvbkbc/linux-xtensa/tree/xtensa-6.4-esp32> [<https://github.com/jcmvbkbc/linux-xtensa/tree/xtensa-6.4-esp32>]
- <https://github.com/jcmvbkbc/binutils-gdb-xtensa/tree/xtensa-2.40-fdpic> [<https://github.com/jcmvbkbc/binutils-gdb-xtensa/tree/xtensa-2.40-fdpic>]
- <https://github.com/jcmvbkbc/gcc-xtensa/tree/xtensa-14-fdpic> [<https://github.com/jcmvbkbc/gcc-xtensa/tree/xtensa-14-fdpic>]
- <https://github.com/jcmvbkbc/uclibc-ng-xtensa/tree/xtensa-fdpic> [<https://github.com/jcmvbkbc/uclibc-ng-xtensa/tree/xtensa-fdpic>]
- <https://github.com/jcmvbkbc/buildroot/tree/xtensa-2023.02-fdpic> [<https://github.com/jcmvbkbc/buildroot/tree/xtensa-2023.02-fdpic>]
- <https://github.com/jcmvbkbc/crostoool-NG/tree/xtensa-fdpic> [<https://github.com/jcmvbkbc/crostoool-NG/tree/xtensa-fdpic>]

Example kernel and rootfs images built from the above sources as well as a toolchain archive: <http://jcmvbkbc.spb.ru/~dumb/tmp/202305161729/> [<http://jcmvbkbc.spb.ru/~dumb/tmp/202305161729/>]

Build toolchain dynconfig library and export XTENSA_GNU_CONFIG for use by the toolchain

```
$ git clone https://github.com/jcmvbkbc/xtensa-dynconfig -b original
$ git clone https://github.com/jcmvbkbc/config-esp32s3 esp32s3
$ make -C xtensa-dynconfig ORIG=1 CONF_DIR='pwd' esp32s3.so
$ export XTENSA_GNU_CONFIG='pwd'/xtensa-dynconfig/esp32s3.so # make sure this environment variable is set for all commands involving building or using the toolchain
```

Build the toolchain

Using crostoool-NG

```
$ git clone https://github.com/jcmvbkbc/crostoool-NG.git -b xtensa-fdpic
$ pushd crostoool-NG
$ ./bootstrap && ./configure --enable-local && make
$ ./ct-ng xtensa-esp32s3-linux-uclibcfdpic
$ CT_PREFIX='pwd'/builds nice ./ct-ng build
$ popd
```

It doesn't complete successfully ATM, failing with the following message, but it builds and installs everything that's important.

```
...
[INFO ] Installing final gcc compiler: done in 572.47s (at 29:47)
[INFO ] =====
[INFO ] Checking dynamic linker symlinks
[EXTRA] Checking dynamic linker for multilib ''
[ERROR] collect2: error: ld returned 1 exit status
[ERROR]
[ERROR] >>
[ERROR] >> Build failed in step 'Checking dynamic linker symlinks'
[ERROR] >> called in step '(top-level)'
[ERROR] >>
[ERROR] >> Error happened in: CT_DoExecLog[scripts/functions@377]
[ERROR] >> called from: CT_FixupLDSO[scripts/functions@1695]
[ERROR] >> called from: CT_IterateMultilibs[scripts/functions@1608]
[ERROR] >> called from: CT_MultilibFixupLDSO[scripts/functions@1761]
[ERROR] >> called from: uClibc-ng_post_cc[scripts/build/libc/uClibc-ng.sh@335]
[ERROR] >> called from: do_libc_post_cc[scripts/build/libc.sh@38]
[ERROR] >> called from: main[scripts/crostoool-NG.sh@697]
[ERROR] >>
...
```

Or manually

```
$ git clone https://github.com/jcmvbkbc/binutils-gdb-xtensa -b xtensa-2.40-fdpic
$ git clone https://github.com/jcmvbkbc/gcc-xtensa -b xtensa-14-fdpic
$ git clone https://github.com/jcmvbkbc/linux-xtensa -b xtensa-6.4-esp32
$ git clone https://github.com/jcmvbkbc/uclibc-ng-xtensa -b xtensa-fdpic
$ mkdir build-xtensa-fdpic-toolchain-esp32s3 ; mkdir build-xtensa-fdpic-nothread-esp32s3
$ ( cd build-xtensa-fdpic-toolchain-esp32s3 ; rm -rf * ; FLAGS_FOR_TARGET='-mtext-section-literals -mfdpic -O2 -g -mforce-l32' nice ../build-xtensa-fdpic-toolchain.sh )
$ ( cd build-xtensa-fdpic-nothread-esp32s3 ; rm -rf * ; CROSS_COMPILE='pwd'../build-xtensa-fdpic-toolchain-esp32s3/root/bin/xtensa-linux-uclibcfdpic- TARGET_CFLAGS='-mauto-lit'
$ ln -s `pwd`/build-xtensa-fdpic-nothread-esp32s3/sysroot build-xtensa-fdpic-toolchain-esp32s3/root/xtensa-linux-uclibcfdpic/
```

build-xtensa-fdpic-toolchain.sh:

```
#!/bin/bash -ex

src_base=$(dirname $(readlink -f "$0"))
binutils_src="$src_base/binutils-gdb"
gcc_src="$src_base/gcc"

target=${TARGET:-xtensa-linux-uclibcfdpic}
base='pwd'

_FLAGS_FOR_TARGET=${FLAGS_FOR_TARGET:-mtext-section-literals -mfdpic -O2 -g}

mkdir binutils && (
  cd binutils
  "$binutils_src/configure" --prefix="$base/root" \
```

```

        --target=$target \
        --disable-shared --disable-werror --disable-gdb --disable-gdbstub \
        CFLAGS='-O0 -g'

    make -j8
    make -j8 install
)

mkdir gcc && (
    cd gcc
    "$gcc_src/configure" --prefix="$base/root" \
        --target=$target \
        --with-sysroot="$base/root/$target/sysroot" \
        --enable-languages=c \
        --disable-shared \
        --enable-__cxa_atexit \
        --enable-tls --disable-threads \
        --without-headers --with-newlib \
        CFLAGS_FOR_TARGET="$_FLAGS_FOR_TARGET" \
        CXXFLAGS_FOR_TARGET="$_FLAGS_FOR_TARGET" \
        CFLAGS='-O0 -g' \
        CXXFLAGS='-O0 -g'

    make -j8 all-gcc
    make -j8 all-target-libgcc
    make -j8 install-gcc
    make -j8 install-target-libgcc
)

build.sh:

#!/bin/bash -ex

if [ $1 = "-r" ]; then
    reconfigure=1
fi
base=$(dirname $(readlink -f "$0"))
export CROSS_COMPILE=${CROSS_COMPILE:-xtensa-dc233c-elf-}
export TARGET_CFLAGS="${TARGET_CFLAGS:-mlongcalls -mauto-litpools -D_LARGEFILE64_SOURCE -D_FILE_OFFSET_BITS=64}"

if [ ! -d build-linux ]; then
    cur=`pwd`
    mkdir -p build-linux
    cd build-linux
    make -C $base/linux ARCH=xtensa O=`pwd` defconfig
    make -C $base/linux ARCH=xtensa O=`pwd` INSTALL_HDR_PATH="$cur" -j8 headers_install
    cd ..
fi

#base=$(dirname "$0")
if [ -z "$reconfigure" ]; then
    [ -f .config ] || make -C $base/uclibc-ng ARCH=xtensa O=`pwd` KERNEL_HEADERS=`pwd`/include UCLIBC_EXTRA_CFLAGS="${TARGET_CFLAGS}" defconfig
else
    make -C $base/uclibc-ng ARCH=xtensa O=`pwd` KERNEL_HEADERS=`pwd`/include UCLIBC_EXTRA_CFLAGS="${TARGET_CFLAGS}" menuconfig
fi

make -C $base/uclibc-ng ARCH=xtensa O=`pwd` KERNEL_HEADERS=`pwd`/include UCLIBC_EXTRA_CFLAGS="${TARGET_CFLAGS}" -j8 "$@"
make -C $base/uclibc-ng ARCH=xtensa O=`pwd` KERNEL_HEADERS=`pwd`/include UCLIBC_EXTRA_CFLAGS="${TARGET_CFLAGS}" DESTDIR=`pwd`/sysroot install

```

Build the rootfs and kernel image

```

$ git clone https://github.com/jcmvbkbc/buildroot -b xtensa-2023.02-fdpic
$ nice make -C buildroot O=`pwd`/build-xtensa-2023.02-fdpic-esp32s3 esp32s3_defconfig
$ nice make -C buildroot O=`pwd`/build-xtensa-2023.02-fdpic-esp32s3 menuconfig # adjust external toolchain location to the one built above
$ nice make -C buildroot O=`pwd`/build-xtensa-2023.02-fdpic-esp32s3

```

Build and flash the bootloader, flash kernel and rootfs images

```

$ git clone https://github.com/jcmvbkbc/esp-idf -b linux-5.0.1
$ pushd esp-idf
$ . export.sh
$ cd examples/get-started/linux_boot
$ idf.py set-target esp32s3
$ idf.py build
$ idf.py flash
$ popd
$ parttool.py write_partition --partition-name linux --input build-xtensa-2023.02-fdpic-esp32s3/images/xipImage
$ parttool.py write_partition --partition-name rootfs --input build-xtensa-2023.02-fdpic-esp32s3/images/rootfs.cramfs

```

The result

```

ESP-ROM:esp32s3-20210327
Build:Mar 27 2021
rst:0x1 (POWERON),boot:0x8 (SPI_FAST_FLASH_BOOT)
SPIWP:0xee
mode:DIO, clock div:1
load:0x3fce3810,len:0x10a0
load:0x403c9700,len:0xa24
load:0x403cc700,len:0x2d04
entry 0x403c988c
I (73) octal_psram: vendor id      : 0x0d (AP)
I (73) octal_psram: dev id        : 0x02 (generation 3)
I (74) octal_psram: density       : 0x03 (64 Mbit)
I (78) octal_psram: good-die      : 0x01 (Pass)
I (83) octal_psram: Latency       : 0x01 (Fixed)
I (89) octal_psram: VCC           : 0x01 (3V)
I (93) octal_psram: SRF           : 0x01 (Fast Refresh)

```

```

I (99) octal_psram: BurstType : 0x01 (Hybrid Wrap)
I (105) octal_psram: BurstLen : 0x01 (32 Byte)
I (110) octal_psram: Readlatency : 0x02 (10 cycles@Fixed)
I (117) octal_psram: DriveStrength: 0x00 (1/1)
I (122) esp_psram: Found 8MB PSRAM device
I (126) esp_psram: Speed: 80MHz
I (130) cpu_start: Pro cpu up.
I (134) cpu_start: Starting app cpu, entry point is 0x40375344
I (0) cpu_start: App cpu up.
I (593) esp_psram: SPI SRAM memory test OK
I (602) cpu_start: Pro cpu start user code
I (602) cpu_start: cpu freq: 160000000 Hz
I (602) cpu_start: Application information:
I (605) cpu_start: Project name: linux_boot
I (610) cpu_start: App version: v5.0.1-4-g680509ab40d1
I (617) cpu_start: Compile time: May 7 2023 16:29:12
I (623) cpu_start: ELF file SHA256: a110e4309915b853...
I (629) cpu_start: ESP-IDF: v5.0.1-4-g680509ab40d1
I (635) cpu_start: Min chip rev: v0.0
I (640) cpu_start: Max chip rev: v0.99
I (644) cpu_start: Chip rev: v0.1
I (649) heap_init: Initializing. RAM available for dynamic allocation:
I (656) heap_init: At 3FC958C0 len 00053E50 (335 KiB): D/IRAM
I (663) heap_init: At 3FCE9710 len 00005724 (21 KiB): STACK/DRAM
I (669) heap_init: At 3FCF0000 len 00008000 (32 KiB): DRAM
I (676) heap_init: At 600FE010 len 00001FF0 (7 KiB): RTCRAM
I (682) esp_psram: Adding pool of 8192K of PSRAM memory to heap allocator
I (690) spi_flash: detected chip: generic
I (694) spi_flash: flash io: dio
I (699) cpu_start: Starting scheduler on PRO CPU.
I (0) cpu_start: Starting scheduler on APP CPU.
I (719) esp_psram: Reserving pool of 32K of internal memory for DMA/internal allocations
ptr = 0x42830000
ptr = 0x42b30000
[ 0.000000] Ignoring boot parameters at (ptrval)
[ 0.000000] Linux version 6.3.0-00022-g5d8354462a70 (jcmvbkbc@octofox) (xtensa-dc233c-elf-gcc (GCC) 13.1.0, GNU ld (GNU Binutils) 2.40) #39 PREEMPT Sun May 7 16:35:44 PDT 2023
[ 0.000000] config ID: c2f0fffe:23090f1f
[ 0.000000] earlycon: esp32uart0 at MMIO32 0x60000000 (options '115200n8')
[ 0.000000] printk: bootconsole [esp32uart0] enabled
[ 0.000000] *****
[ 0.000000] ** NOTICE NOTICE NOTICE NOTICE NOTICE NOTICE **
[ 0.000000] **
[ 0.000000] ** This system shows unhashed kernel memory addresses **
[ 0.000000] ** via the console, logs, and other interfaces. This **
[ 0.000000] ** might reduce the security of your system. **
[ 0.000000] **
[ 0.000000] ** If you see this message and you are not debugging **
[ 0.000000] ** the kernel, report this immediately to your system **
[ 0.000000] ** administrator! **
[ 0.000000] **
[ 0.000000] ** NOTICE NOTICE NOTICE NOTICE NOTICE NOTICE **
[ 0.000000] *****
[ 0.000000] Zone ranges:
[ 0.000000] Normal [mem 0x000000003c030000-0x000000003c82ffff]
[ 0.000000] Movable zone start for each node
[ 0.000000] Early memory node ranges
[ 0.000000] node 0: [mem 0x000000003c030000-0x000000003c82ffff]
[ 0.000000] Initmem setup node 0 [mem 0x000000003c030000-0x000000003c82ffff]
[ 0.000000] pcpu-alloc: s0 r0 d32768 u32768 alloc=1*32768
[ 0.000000] pcpu-alloc: [0] 0
[ 0.000000] Built 1 zonelists, mobility grouping off. Total pages: 2032
[ 0.000000] Kernel command line: earlycon=esp32uart,mmio32,0x60000000,115200n8 console=ttyS0,115200n8 debug rw root=mtd:data no_hash_pointers
[ 0.000000] Dentry cache hash table entries: 1024 (order: 0, 4096 bytes, linear)
[ 0.000000] Inode-cache hash table entries: 1024 (order: 0, 4096 bytes, linear)
[ 0.000000] mem auto-init: stack:off, heap alloc:off, heap free:off
[ 0.000000] virtual kernel memory layout:
[ 0.000000] lowmem : 0x3c030000 - 0x3c830000 ( 8 MB)
[ 0.000000] .text : 0x42830000 - 0x429e0d28 ( 1731 kB)
[ 0.000000] .rodata : 0x429e1000 - 0x42a1f000 ( 248 kB)
[ 0.000000] .data : 0x3c030000 - 0x3c0a9420 ( 485 kB)
[ 0.000000] .init : 0x3c0a9420 - 0x3c0adf00 ( 18 kB)
[ 0.000000] .bss : 0x3c0adf00 - 0x3c0e1988 ( 206 kB)
[ 0.000000] Memory: 7332K/8192K available (1731K kernel code, 485K rwdma, 248K rodata, 88K init, 206K bss, 860K reserved, 0K cma-reserved)
[ 0.000000] SLUB: Hwalign=16, Order=0-3, MinObjects=0, CPUs=1, Nodes=1
[ 0.000000] rcu: Preemptible hierarchical RCU implementation.
[ 0.000000] rcu: RCU calculated value of scheduler-enlistment delay is 10 jiffies.
[ 0.000000] NR_IRQS: 33
[ 0.000000] rcu: srcu_init: Setting srcu_struct sizes based on contention.
[ 0.000000] clocksource: ccount: mask: 0xffffffff max_cycles: 0xffffffff, max_idle_ns: 11945377789 ns
[ 0.000086] sched_clock: 32 bits at 160MHz, resolution 6ns, wraps every 13421772796ns
[ 0.008110] Calibrating delay loop (skipped)... 160.00 BogoMIPS preset
[ 0.014370] pid_max: default: 4096 minimum: 301
[ 0.021337] Mount-cache hash table entries: 1024 (order: 0, 4096 bytes, linear)
[ 0.026541] Mountpoint-cache hash table entries: 1024 (order: 0, 4096 bytes, linear)
[ 0.077044] rcu: Hierarchical SRCU implementation.
[ 0.077673] rcu: Max phase no-delay instances is 1000.
[ 0.091972] devtmpfs: initialized
[ 0.115533] clocksource: jiffies: mask: 0xffffffff max_cycles: 0xffffffff, max_idle_ns: 1911260446275000 ns
[ 0.116804] futex hash table entries: 16 (order: -5, 192 bytes, linear)
[ 0.139676] NET: Registered PF_NETLINK/PF_ROUTE protocol family
[ 0.156790] platform soc: Fixed dependency cycle(s) with /soc/intc@600c2000
[ 0.219233] clocksource: Switched to clocksource ccount
[ 0.266507] NET: Registered PF_INET protocol family
[ 0.274408] IP idents hash table entries: 2048 (order: 2, 16384 bytes, linear)
[ 0.294920] tcp_listen_portaddr_hash hash table entries: 1024 (order: 0, 4096 bytes, linear)
[ 0.296128] Table-perturb hash table entries: 65536 (order: 6, 262144 bytes, linear)
[ 0.303770] TCP established hash table entries: 1024 (order: 0, 4096 bytes, linear)
[ 0.309855] TCP bind hash table entries: 1024 (order: 1, 8192 bytes, linear)
[ 0.315629] TCP: Hash tables configured (established 1024 bind 1024)

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[ 0.326107] UDP hash table entries: 256 (order: 0, 4096 bytes, linear)
[ 0.329877] UDP-Lite hash table entries: 256 (order: 0, 4096 bytes, linear)
[ 0.340029] NET: Registered PF_UNIX/PF_LOCAL protocol family
[ 0.361833] workingset: timestamp_bits=30 max_order=11 bucket_order=0
[ 2.712542] 60000000.serial: ttyS0 at MMIO 0x60000000 (irq = 1, base_baud = 0) is a ESP32 UART
[ 2.714408] printk: console [ttyS0] enabled
[ 2.714408] printk: console [ttyS0] enabled
[ 2.720147] printk: bootconsole [esp32uart0] disabled
[ 2.720147] printk: bootconsole [esp32uart0] disabled
[ 2.759786] physmap-flash 42830000.flash: physmap platform flash device: [mem 0x42830000-0x4302ffff]
[ 2.761481] 2 fixed-partitions partitions found on MTD device 42830000.flash
[ 2.765620] Creating 2 MTD partitions on "42830000.flash":
[ 2.772746] 0x000000000000-0x0000000300000 : "linux"
[ 2.787049] 0x000000300000-0x0000000880000 : "data"
[ 2.787809] mtd: partition "data" extends beyond the end of device "42830000.flash" -- size truncated to 0x500000
[ 2.814687] NET: Registered PF_PACKET protocol family
[ 3.044471] cramfs: checking physical address 0x42b30000 for linear cramfs image
[ 3.045351] cramfs: linear cramfs image on mtd:data appears to be 1808 KB in size
[ 3.052366] VFS: Mounted root (cramfs filesystem) readonly on device 31:1.
[ 3.057890] devtmpfs: mounted
[ 3.062212] Freeing unused kernel image (initmem) memory: 12K
[ 3.064482] This architecture does not have kernel memory protection.
[ 3.072536] Run /sbin/init as init process
[ 3.074991]   with arguments:
[ 3.077870]     /sbin/init
[ 3.081602]   with environment:
[ 3.083725]     HOME=/
[ 3.086028]     TERM=linux
Starting syslogd: OK
Starting klogd: OK
Running sysctl: OK
seedrng: can't create directory '/var/lib/seedrng': Read-only file system
Starting network: OK

Welcome to Buildroot
buildroot login: root
~ # cat /proc/cpuinfo
CPU count      : 1
CPU list       : 0
vendor_id      : Tensilica
model          : Xtensa LX7.0.12
core ID        : LX7_ESP32_S3_MP
build ID       : 0x90f1f
config ID      : c2f0ffff:23090f1f
byte order     : little
cpu MHz        : 160.00
bogomips       : 320.00
flags          : nmi debug ocd density boolean loop nsa minmax sext clamps mac16 mul16 mul32 mul32h fpu s32c1i
physical aregs : 64
misc regs      : 4
ibreak         : 2
dbreak         : 2
num ints       : 32
ext ints       : 26
int levels     : 6
timers         : 3
debug level    : 6
icache line size: 4
icache ways    : 1
icache size    : 0
icache flags   :
dcache line size: 16
dcache ways    : 1
dcache size    : 0
dcache flags   :
~ # free

```

	total	used	free	shared	buff/cache	available
Mem:	7344	3264	3444	0	636	3208

```
~ #
```